

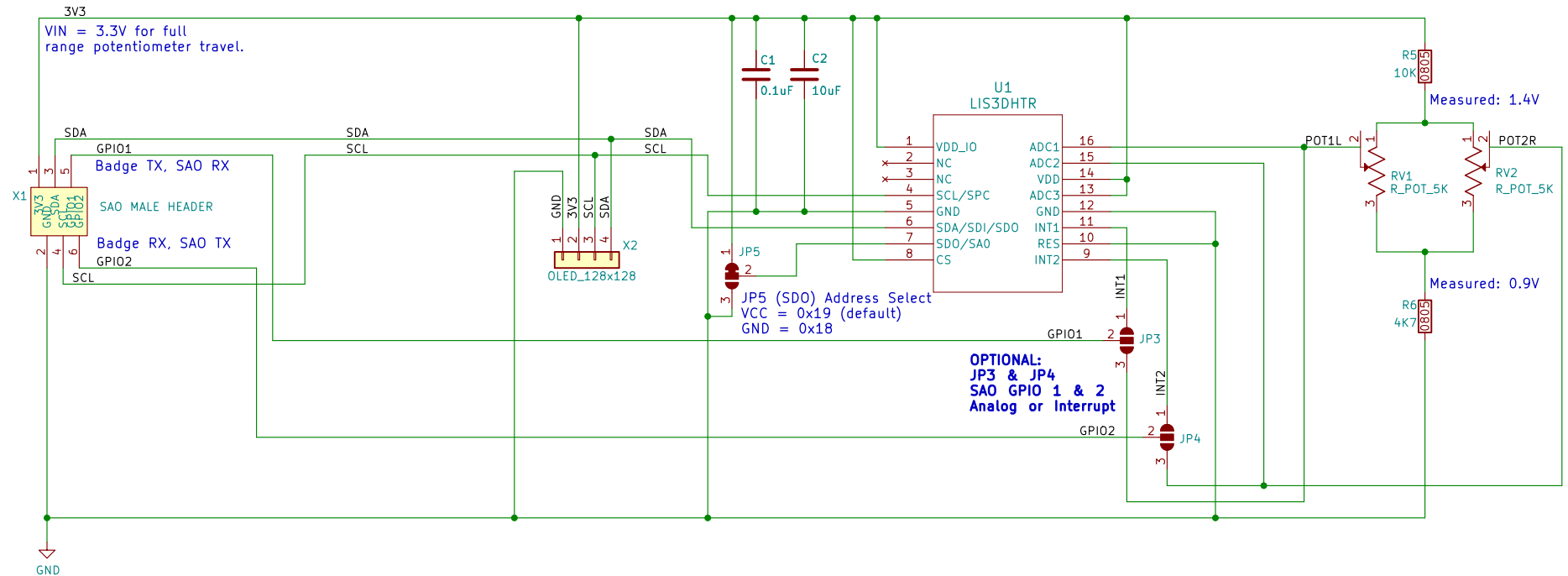
SAO SKETCH – OLED VERSION 1.4

SAO PORT

DISPLAY

ACCELEROMETER

ANALOG KNOBS



SAO PORT (6-pin, 2x3, 0.1 inch spacing, male pins)
Used AMP 68691-406HLF, 2x3 header 0.1 inch, 0.538" (13.67mm) OAL instead of the standard:
<https://hackaday.io/project/175182-simple-add-ons-sao>
Adam Tech BHR-06-VUA male, polarized header
<https://www.digikey.com/en/products/detail/adam-tech/BHR-06-VUA/10414837>

OLED DISPLAY (Address 0x3C, 128 x 128 pixel, 1.5 inch, 16 shade grayscale, SSD1327 controller)
<https://learn.adafruit.com/adafruit-grayscale-1-5-128x128-oled-display>
Optional address 0x3D selectable by moving resistor in upper left on the back of the blue OLED board.
MFR'd <https://goldenmorninglcd.com/oled-display-module/1.5-inch-128x128-ssd1327-gme128128-02/>

ACCELEROMETER (Address 0x19, STM LIS3DHTR, 3-axis)
<https://learn.sparkfun.com/tutorials/lis3dh-hookup-guide/all>
CS is high for I2C mode.
ADC theoretical input range is 0.8 to 1.6 V (aka 1.2 +/- .4)
ADC 1 and 2 used for potentiometers. ADC3 for internal temperature or 3V3 rail voltage calibration.
Optional address 0x18 if JP5 (SAO high) is changed to (SAO low).

OPTIONAL JP3 & JP4 CONFIG
Solder pads 1-2 to connect accelerometer interrupt output pins to SAO GPIO 1&2.
Solder pads 2-3 to connect analog potentiometer outputs to SAO GPIO 1&2

POTENTIOMETERS/ANALOG KNOBS (Accessed through accelerometer)
Left pot connected to accelerometer input ADC1.
Right pot connected to accelerometer input ADC2.

ERRATA 2026-09-08

Connected ADC3 to the 3V3 rail, forgot it can't read above 1.4V, and it needs a voltage divider to be useful. So it's not useful in this version.

All non-polarized capacitors are X7R or X5R ceramic unless otherwise noted.
https://github.com/ageppert/SAO_Etch_sAo_Sketch/
<https://hackaday.io/project/197581-etch-sao-sketch>

Concept and design by Andy Geppert © www.MachineIdeas.com

Sheet: /
File: SAO_Sketch.kicad_sch

Title: Etch sAo Sketch – OLED Version

Size: A Date: 2026-08-23

Rev: 1.4

KiCad E.D.A. 10.0.3

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